Lakshya Agarwal

Jaipur, India | la9928609007@gmail.com | lakshya9461.in | github.com/Lakshya9461 | linkedin.com/in/lakshya70

Summary

Computer Science undergraduate focused on AI/ML, software engineering, and self-hosted systems infrastructure. Experienced building clinical ML pipelines, local LLM/agent infrastructure, and privacy-focused applications across Linux, Docker, and networked environments.

Education

JK Lakshmipat University — Jaipur, India *Aug 2024 – May 2028*

B.Tech in Computer Science *50% merit scholarship, R-CAT Quiz-A-Thon programme*

Relevant coursework: Data Structures & Algorithms, Machine Learning, Computer Networks, Database Management Systems, Cloud Computing, System Design

Research Experience

Undergraduate Research Intern — Early Sepsis Prediction, MIMIC-IV *2026 – Present*

Supervised by Dr. Amit Sinhal, JK Lakshmipat University

- Built an ML pipeline for 6-hour-horizon Sepsis-3 onset prediction on MIMIC-IV ICU data, including suspected-infection identification, patient/admission/ICU-stay preprocessing, and temporal feature construction.
- Implemented and benchmarked Logistic Regression, XGBoost, and an LSTM sequence model, using GroupKFold cross-validation to control patient-level data leakage; evaluated with AUROC, AUPRC, sensitivity, specificity, and F1.
- Best model (LSTM) reached AUROC 0.779 / AUPRC 0.394, outperforming XGBoost and Logistic Regression baselines; currently investigating external validation on eICU / HiRID / AmsterdamUMCdb.

Projects

Local LLM & AI Agent Infrastructure Ollama, WSL2, Nginx, VS Code Continue, Hermes Agent

- Administer a remote GPU workstation (RTX 4070) as a local model server, running Ollama with Qwen, Gemma, DeepSeek, and GPT-OSS models behind an Nginx reverse proxy managed via WinSW.
- Configured OpenAI-compatible local endpoints for coding/agent tools (Continue, OpenCode) and run Hermes Agent in WSL2 with isolated profiles for coding, research, and daily-use workflows.
- Provisioned isolated per-user RDP access and diagnosed/resolved recurring GPU driver and stability failures caused by Defender scanning and gateway processes.

Self-Hosted Home Server Debian Linux, Docker, Cloudflare Tunnel, Tailscale

- Designed and maintain a self-hosted homelab running containerized media, file-sharing, and utility services on Debian Linux, with automated Docker/Git deployment and remote access via Cloudflare Tunnel and Tailscale (no exposed public ports).

Aegis Browser Electron, Chromium, JavaScript

- Built a privacy-focused desktop browser using secure Electron architecture (context isolation, preload bridges) with a 500+ domain ad-blocking/anti-tracking pipeline, incognito sessions, and tab suspension.

Technical Skills

Programming: Python, SQL, JavaScript, C

AI / ML: PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Time-Series Modeling, LLMs, Prompt Engineering

AI / LLM Infrastructure: Ollama, OpenAI-compatible APIs, Local LLM Inference, OpenCode, Hermes Agent

Software Development: Flutter, Electron, REST APIs, SQLite, Git, Clean Architecture

Systems / Infrastructure: Linux (Debian/Ubuntu), Windows, WSL, Docker, Docker Compose, Nginx, Cloudflare Tunnel, Tailscale, SSH, Linux Networking

Databases: SQL, SQLite, MySQL

Experience

Tech & AI Writer — Freelance *Feb 2026 – Present*

- Publish technical articles on Linux, self-hosting, and AI tooling, including *"I Ran 4 Operating Systems on a 4 GB Home Server"* and *"30 Days Replacing Windows with Linux."*